

IPv4 Addressing Specifics

Location HQ-TonontoBR – Oshawa	End Devices	Network Address/Subnet Mask	First Host Ip Address	Last Host Ip Address	Broadcast Address	Host Space
HQ-Subnet 1- Management Devices	12	172.17.5.0 255.255.255.240/28	172.17.5.1	172.17.5.14	172.17.5.15	Number of Usable Hosts: 14 End Devices: 12
HQ-Subnet 2- Corporate Workstations	124	172.17.5.16 255.255.255.128/25	172.17.5.17	172.17.5.142	172.17.5.143	Number of Usable Hosts: 126 End Devices: 124
HQ-Subnet 3- Printers	14	172.17.5.144 255.255.255.240/28	172.17.5.145	172.17.5.158	172.17.5.159	Number of Usable Hosts: 14 End Devices: 14
HQ-Subnet 4- Servers	28	172.17.5.160 255.255.255.224/27	172.17.5.161	172.17.5.190	172.17.5.191	Number of Usable Hosts: 32 End Devices: 28
BR-Subnet 1- Management Devices	25	172.17.5.192 255.255.255.224/27	172.17.5.193	172.17.5.222	172.17.5.223	Number of Usable Hosts: 32 End Devices: 25
BR-Subnet 3- Printers	30	172.17.5.224 255.255.255.224/27	172.17.6.225	172.17.6.254	172.17.6.255	Number of Usable Hosts: 32 End Devices: 30
BR-Subnet 2- Corporate Workstations	120	172.17.6.0 255.255.255.128/25	172.17.6.1	172.17.6.126	172.17.6.127	Number of Usable Hosts: 128 End Devices: 120
BR-Subnet 4- HR+BR VPN Workstations	125	172.17.6.128 255.255.255.128/25	172.17.7.1	172.17.7.254	172.17.7.255	Number of Usable Hosts: 126 End Devices: 125

Brief Rationale

The initial address space allocated to our group, group number 5, is 172.17.5.0 with a /23 CIDR. The subnetting for SecureArc consists of a variable length subnetting masking with addresses in the range of 172.17.5.0 to 172.17.5.255, then 172.17.6.0 to 172.6.255. The subnets were reordered to better fit the subnets, such that BR-Subnet 3 fits in the 172.17.5.0 to 172.17.5.255 range along with HQ-Subnets 1 to 4 and BR-Subnet 1, but BR-Subnet 2 does not fit. This is due to Br-Subnet 2 having 120 end devices but BR-Subnet 3 only has 30, and with HQ-Subnets 1 to 4 and BR-Subnet 1, there would only be 32 address spaces left.